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Real-Time sEMG Control of a Bionic Hand for Educational Demonstration

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Abstract

This thesis presents the design and evaluation of a 3D-printed, EMG-controlled bionic arm. The system integrates mechanical fabrication, surface electromyography (sEMG) signal processing and real-time finger actuation.

The mechanical structure, based on the open-source InMoov project, was modified for durability and reduced friction by adding Teflon tubing at tendon contact points, improving tendon anchoring and creating a custom stand for electronics. sEMG signals from the flexor carpi radialis were filtered using a 6th-order Butterworth bandpass filter (65-135 Hz). Three features, root mean square (RMS), variance (VAR) and waveform length (WL), were evaluated for accuracy and detection delay. RMS and WL outperformed VAR, achieving $\approx 95\%$ accuracy at a 90 ms window size with low delay.

A dual-threshold control algorithm mitigated noise spikes from electrode disconnection, disabling actuation and providing a visual alert. Real-time tests with four subjects showed RMS and WL provided comparable, reliable control. Future work should investigate long-term use and automatic recalibration to improve robustness.

Kurzfassung

Diese Arbeit beschreibt die Entwicklung und Evaluierung eines 3D-gedruckten, EMG-gesteuerten bionischen Arms. Das System kombiniert mechanische Fertigung, die Verarbeitung von Oberflächen-EMG-Signalen (sEMG) und die Echtzeitsteuerung der Finger.

Die auf dem Open-Source-Projekt InMoov basierende Mechanik wurde durch Teflonröhrchen an Sehnenkontaktpunkten, verbesserte Sehnenbefestigungen und einen maßgefertigten Stand für die Elektronik optimiert. sEMG-Signale des Musculus flexor carpi radialis wurden mit einem 6.-Ordnungs-Butterworth-Bandpassfilter (65-135 Hz) gefiltert. Drei Merkmale - Root Mean Square (RMS), Varianz (VAR) und Wellenlänge (WL) - wurden hinsichtlich Genauigkeit und Erkennungsverzögerung bewertet. RMS und WL übertrafen VAR und erreichten bei einer Fenstergröße von 90 ms eine Genauigkeit von 95% bei niedriger Verzögerung.

Ein Dual-Schwellenwert-Algorithmus verhinderte Fehlaktivierungen durch Elektrodenunterbrechungen, deaktivierte die Ansteuerung und löste eine visuelle Warnung aus. Echtzeittests mit vier Probanden zeigten, dass RMS und WL vergleichbare und zuverlässige Steuerung ermöglichten. Zukünftige Arbeiten sollten Langzeittests, hybride Merkmalsansätze und eine automatische Neukalibrierung untersuchen, um die Robustheit zu erhöhen.

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1 Introduction

1.1 Motivation

In March 2022, the Graz University of Technology (TU Graz) founded the MINKT laboratory at Campus Infeldgasse, a dedicated space for fostering scientific curiosity and hands-on learning among young people. This laboratory offers more than twenty interactive experiments, primarily designed for children and teenagers, many of which are created in collaboration with the university's various departments and institutes. The environment is crafted to encourage self-driven exploration, allowing visitors to engage with scientific concepts at their own pace and according to their interests. Supervisors are present to support learning and answer questions that arise during these explorations.

Over two years of working in the MINKT laboratory, I observed how interactive experiments inspire genuine excitement among participants. Reflecting on my own experience, how building and experimenting sparked my interest in science and engineering, motivated me to develop a new experiment rooted in my current field of study. This project aims to expand the laboratory's curriculum and introduce the fundamental principles of biomedical engineering in an accessible way.

1.2 Problem Statement

Despite the MINKT laboratory's broad selection of STEM-based projects, a dedicated experiment showcasing the field of biomedical engineering is currently missing. This creates a notable gap in the exposure offered to young visitors. While children and teenagers are introduced to principles of physics, chemistry, computer science and general engineering through existing experiments, biomedical engineering, a field that combines biology, medicine and engineering to solve real-world healthcare challenges, is missing.

It was decided to develop an experiment that allows children to learn about biomedical engineering in an interactive way. To gather ideas for the experiment, all institutes related to biomedical engineering at the university were contacted. Based on the suggestions received, the experiment was designed around the control of a bionic arm using real-time electrophysiological data like electrocardiography or electromyography. A bionic arm illustrates biomedical engineering principles by showing how signals from the human body can be measured, processed and used to control assistive technology in real time. However, designing such a system for educational use presents unique challenges: it must be durable, intuitive, low-maintenance and safe for casual use by children with minimal supervision. Addressing these design challenges is the central focus of this thesis.

1.3 Project Goals and Objectives

The primary goal of this thesis is to design and implement a functional prototype of a bionic arm at the MINKT laboratory, which visitors can control through their own physiological signals. Achieving this involves the following objectives:

- Developing a modular and repairable bionic hand model suitable for interactive educational use.
- Evaluating and selecting appropriate electrophysiological signals that can be used easily, safely and reliably to control the bionic arm within the environment of the MINKT laboratory.
- Designing and building a robust sensor system that effectively captures and processes the user’s physiological signals to generate accurate motor control commands.
- Testing and validating all system components to ensure safety, reliability and educational value, providing clear documentation for future development and use by laboratory engineers.

The solution must be user-friendly, cost-effective and designed for minimal supervision so that children can safely explore the experiment independently.

1.4 Thesis Scope and Structure

This thesis documents the development of the sEMG-controlled bionic hand prototype. The scope encompasses the replication of the InMoov hand hardware and the setup of a functional sEMG acquisition chain. It includes the implementation and comparison of selected sEMG onset detection algorithms, the integration of signal processing with servo motor control and basic functional testing results.

The remainder of this thesis is structured as follows. Section 2 provides theoretical background and reviews relevant literature. This covers BME principles, sEMG signal characteristics, acquisition, processing and open-source hardware. Section 3 details the specific methodology used for system design and implementation. It describes hardware construction, software development, system integration and testing protocols. Section 4 presents the outcomes of the project. This includes the built hardware, signal acquisition results, algorithm performance and overall system functionality. Section 5 interprets these results, discusses challenges, limitations and suggests future work. Finally, Section 6 summarizes the key findings and contributions of the thesis.

2 Background and Literature Review

2.1 Electrophysiological Signals

Electrophysiological signals are electrical activities produced by excitable cells such as neurons, muscle fibers and heart tissue. These signals, measured as voltage differences on or within the body, are essential for studying neuromuscular function, medical diagnostics, rehabilitation and human-machine interaction. Surface-recorded electrophysiological signals most frequently utilized are electroencephalogram, electrooculogram, electrocardiogram and electromyogram. Each of these methods provides unique information that can be applied to different medical and engineering purposes.

EEG measures brain activity using typically 21 to 256 electrodes placed on the scalp. It records the combined electrical signals from groups of neurons, producing waveforms that reflect states such as alertness, relaxation and sleep. EEG is commonly used to diagnose epilepsy, sleep disorders and brain injuries, as well as in brain-computer interface (BCI) systems. It is non-invasive and offers high temporal resolution, making it ideal for tracking brain function.

EOG records the electrical potential created by eye movements. Electrodes placed around the eyes detect changes in the corneo-retinal potential as the eyes move. EOG is widely used for studying visual behavior, controlling assistive technologies and monitoring sleep stages, particularly REM. It also plays a role in diagnosing eye disorders and assessing balance and vestibular function.

ECG measures the heart's electrical signals during each heartbeat. Electrodes placed on the chest and limbs record the timing and strength of these signals. The resulting waveforms provide crucial information about heart rhythm, rate and conduction. ECG is a key diagnostic tool for detecting arrhythmias, heart attacks and other cardiovascular conditions due to its simplicity and reliability.

EMG records electrical activity in skeletal muscles during contraction. Using surface or needle electrodes, it measures motor unit action potentials to assess muscle performance and nerve health. EMG is used to diagnose conditions such as muscular dystrophy and nerve compression. It is also applied in rehabilitation, ergonomics and in controlling prosthetic devices.

Among these four methods, surface EMG (sEMG) is the most suitable for this project. EEG and EOG require complex setups which would be too time consuming for a short experiment, while ECG is not appropriate because heart rate cannot be precisely controlled to operate a hand. Therefore, sEMG has been chosen as the input signal to control the hand in this project.

2.2 Surface Electromyography (sEMG)

sEMG measures composite Motor Unit Action Potentials (MUAPs) in superficial muscles and is particularly favored due to its non-invasiveness, repeatability and broad applications in sports science, rehabilitation and bioengineering. sEMG signals originate primarily within 5-10 mm beneath the skin, reflecting the spatial and temporal summation of MUAPs. Typically, sEMG signals range from a few microvolts to 10-15 millivolts, with dominant frequency components between 10 and 500 Hz. These signals convey physiological information on neural drive, muscle recruitment, fatigue and force exertion, although distinguishing these parameters individually is challenging. Due to its superficial detection volume, sEMG is limited in capturing deep muscle activity. Applications include prosthetic and exoskeleton control, diagnostics, motion analysis, ergonomics, biofeedback and sports analytics [1], [2], [3].

Technical considerations for sEMG involve selecting appropriate electrodes, with Ag/AgCl electrodes being the gold standard due to their low impedance and stable skin interface. Electrode characteristics such as geometry, diameter (3-5 mm), placement, inter-electrode distance (5-10 mm) and thorough skin preparation significantly influence signal quality, noise and susceptibility to artifacts. Common detection configurations include monopolar, bipolar (differential) and high-density arrays. Bipolar configurations, capturing differences between adjacent electrodes, reduce common-mode noise and physiological artifacts, making them suitable for most applications. High-density arrays further enhance spatial localization and muscle activity mapping. Major noise sources affecting sEMG include powerline interference, motion artifacts, muscle crosstalk and electronics-induced noise. Minimizing these issues requires proper hardware with high input impedance amplifiers, strong common-mode rejection ratios (CMRR), good skin-electrode contact and effective pre-processing techniques [1], [4], [5], [6].

2.3 sEMG: Characteristics and Challenges

The characteristics of sEMG signals are influenced by physiological factors such as motor unit recruitment, firing rates, muscle fiber conduction velocities and anatomical placement of the electrodes [6]. According to the size principle, larger motor units are recruited as contraction intensity increases, leading to a corresponding rise in signal amplitude. Additionally, the frequency content of the signal is modulated by contraction intensity, muscle fatigue and fiber type composition [4], [7].

Accurate sEMG acquisition depends significantly on the detection hardware. Surface electrodes, typically made of Ag/AgCl, must maintain optimal skin contact to minimize impedance and ensure high-fidelity recordings [4]. Skin preparation, electrode type, placement and fixation are essential to reduce variability and motion artifacts. The amplifiers used in sEMG systems should exhibit high input impedance ($>100\text{ M}\Omega$) and common-mode rejection ratio ($\text{CMRR} \geq 90\text{ dB}$) to prevent signal loss in case of high source impedances [6].

One of the primary challenges in sEMG acquisition is the contamination of the signal by various types of noise and artifacts. These include powerline interference (typically 50/60 Hz), motion artifacts (10-20 Hz), baseline noise and physiological interference such as electrocardiographic (ECG) signals [6]. Each contaminant has distinct frequency characteristics and may alter both time-domain and spectral features of the signal. For instance, spectral deformation caused by interference can lead to inaccurate muscle fatigue assessments or prosthetic misclassification.

To address these issues, a combination of filtering strategies is employed. Low-pass filters help eliminate high-frequency interferences, while band-pass filters (typically 10-500 Hz) isolate the physiologically relevant bandwidth of sEMG [6]. Notch filters are used to attenuate narrowband deterministic interference such as powerline interference without distorting the signal of interest. Filter design plays a pivotal role: Butterworth filters offer smooth roll-off with minimal distortion, FIR filters ensure linear phase response and Chebyshev or elliptic filters provide sharper cutoffs at the cost of potential phase distortion [7].

Despite rigorous filtering and best practices in hardware setup, sEMG signals often retain residual noise. Advanced strategies such as biosignal quality assessment and contamination classification are increasingly used to detect, identify and mitigate specific artifacts [6]. Such methods contribute to maintaining high signal integrity, which is essential for downstream applications including onset detection, prosthetic control and muscle health diagnostics.

2.4 Filtering of sEMG Signals

Analog and digital filters are used during preprocessing to enhance signal quality by attenuating artifacts and noise. Both types offer distinct characteristics and trade-offs. Analog filters operate on continuous-time signals using physical components such as resistors and capacitors, making them suitable for real-time analog front-end systems and for effective suppression of radio-frequency interferences. However, their performance is limited by component tolerances, thermal drift and environmental interference [7].

Digital filters, by contrast, process discretized signals through mathematical operations on sampled data, enabling greater flexibility, precision and adaptability. They are particularly suitable for biomedical signal processing, as they can be modified in software, support complex filter designs and are unaffected by hardware variability. However, they introduce latency due to analog-to-digital (ADC) and digital-to-analog (DAC) conversions, as well as computational delays [8].

Given the software-based nature and real-time requirements of this project, digital filters were chosen for their ease of implementation, cost-effectiveness and adaptability. They can emulate analog filters by transforming classical analog transfer functions, preserving essential characteristics such as cutoff frequencies and passband flatness. This enables

the suppression of specific noise sources or unwanted frequency components through targeted filtering.

Four fundamental filter types are typically used in sEMG signal processing:

- **Analog Low-pass filters** suppress frequencies above a certain cutoff, removing high-frequency noise and aliasing effects prior to digitization.
- **Analog High-pass filters** eliminate low-frequency noise such as baseline drift or motion artifacts.
- **Notch filters** are designed to attenuate a narrow band of frequencies, commonly centered at 50/60 Hz to remove power line interference.
- **Band-pass filters** allow only a defined frequency band to pass through, typically set between 20 Hz and 500 Hz for sEMG applications.

2.5 Digital Butterworth IIR Filters

Digital filters are generally implemented as either *Finite Impulse Response (FIR)* or *Infinite Impulse Response (IIR)* systems. FIR filters are non-recursive and inherently stable, offering linear phase response, which is beneficial when phase accuracy is critical. However, achieving sharp filter transitions with FIR designs requires long filter lengths, increasing computational complexity. In contrast, IIR filters use a recursive structure, enabling a steeper roll-off with fewer coefficients and thus lower computational load. The trade-off is a potential risk of instability and phase distortion [8].

For this project, IIR filters were selected due to their computational efficiency, which is advantageous in real-time sEMG applications. The general form of an IIR filter is expressed as:

$$y_n = \sum_{k=0}^K c_k x_{n-k} + \sum_{k=1}^K d_k y_{n-k} \quad (1)$$

where the output y_n is computed as a function of both current and past input values x_{n-k} , as well as past output values y_{n-k} . This recursive dependency enables the realization of complex filter responses using relatively few operations.

To design IIR filters, analog prototypes such as the Butterworth filter are commonly employed. The Butterworth filter is characterized by a maximally flat magnitude response in the passband, ensuring smooth signal attenuation without ripples. Its frequency response for a low-pass filter of order N is given by:

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{j\omega}{\omega_c}\right)^{2N}} \quad (2)$$

where ω_c is the cutoff frequency and N denotes the filter order. Higher-order filters exhibit steeper transition bands and better selectivity, but also pose numerical challenges when implemented directly in digital form, such as coefficient instability and quantization errors.

To mitigate such instability, filters are often implemented as cascaded second-order sections, meaning the overall filter is constructed by chaining together multiple simpler second-order filters, each responsible for a subset of the total frequency response. This modular structure enhances numerical stability and computational precision, particularly in systems with limited resolution or fixed-point arithmetic, which are common in embedded biomedical devices.

Digital realization of analog filters is typically achieved using the bilinear transformation, which maps the continuous s -domain into the discrete z -domain through a non-linear frequency warping:

$$s = \frac{2}{T} \cdot \frac{z - 1}{z + 1} \quad (3)$$

This transformation preserves filter stability and maps the entire analog frequency range onto the digital frequency axis in a one-to-one fashion. The relationship between analog and digital frequency components is described by:

$$\omega_a = \tan\left(\frac{\omega_d \cdot T}{2}\right) \quad (4)$$

By applying the bilinear transformation to a normalized analog prototype, digital low-pass and high-pass filters can be derived with predictable behavior. For instance, a first-order analog high-pass Butterworth filter with the transfer function

$$G(s) = \frac{\omega_c}{s + \omega_c} \quad (5)$$

can be discretized to yield a digital filter suitable for real-time signal conditioning applications. The resulting difference equations are implemented iteratively in software, allowing continuous adaptation of the filter output as new samples arrive.

In conclusion, digital Butterworth IIR filters provide an effective and practical solution for real-time sEMG filtering due to their balance between frequency selectivity, numerical stability and computational efficiency. Proper analog-to-digital transformation ensures robust filtering behavior, while also attenuating unwanted noise components. As noted in prior studies [7], [8], filtering remains a critical step in preparing sEMG signals for downstream processing such as feature extraction and onset detection, with direct implications for system reliability, responsiveness and accuracy.

2.6 Feature Extraction and Onset Detection in sEMG Signals

Feature extraction refers to the process of deriving informative and discriminative attributes from raw signal data within fixed-duration windows. These features encapsulate salient temporal, spectral, or nonlinear characteristics of the signal and serve to reduce dimensionality while enhancing class separability.

Phinyomark et al. [9] presented a comprehensive taxonomy of feature extraction techniques employed in myoelectric pattern recognition systems, categorizing them into time-domain, frequency-domain, time–frequency-domain and nonlinear groups. Within the time-domain category, most suitable for real-time processing, they emphasized features such as mean absolute value (MAV), waveform length (WL), zero crossings (ZC) and slope sign changes (SSC), owing to their low computational cost and high discriminative power. Furthermore, the authors advocated for dimensionality reduction techniques, including principal component analysis (PCA), to minimize redundancy and improve system generalizability.

Focusing more specifically on onset detection, Cho et al. [10] conducted an empirical evaluation of nine commonly used time-domain features for sEMG-based movement onset detection. To address the limitations of conventional binary classification metrics, the authors proposed a novel evaluation framework incorporating three temporal performance criteria: average detection delay, false detection ratio (FDR) and probability of detection failure. Their findings revealed a consistent trade-off between responsiveness and precision. For example, waveform length (WL) achieved the shortest detection delay, occasionally even preceding the annotated onset, yet exhibited a substantially higher false detection rate. In contrast, features such as variance (VAR) and root mean square (RMS) demonstrated superior precision, although with increased detection latency.

Both studies underscore the critical influence of sliding window parameters on detection performance. Phinyomark et al. [9] noted that smaller window sizes enhance responsiveness but are more susceptible to noise, whereas larger windows improve stability at the expense of temporal resolution. Cho et al. [10] corroborated this observation, further identifying RMS and modified MAV (MMAV2) as particularly robust features across varying window lengths.

To further optimize detection performance, Cho et al. [10] introduced a label-shifting strategy during training. By systematically adjusting the timing of ground truth onset labels, they demonstrated that a negative shift (earlier onset) improved responsiveness but increased false positives, whereas a positive shift enhanced precision by suppressing early detections. This method offers a promising avenue for fine-tuning the trade-off between detection speed and reliability.

In summary, the selection of appropriate features, the tuning of windowing parameters and the implementation of adaptive training strategies are all crucial for achieving robust and timely onset detection in sEMG-based systems.

3 Method

3.1 3D Printing

3.1.1 Printer

Two Fused Deposition Modeling (FDM) 3D printers were available at the MINKT laboratory: the Snapmaker J1S and the Snapmaker Artisan. The Artisan is a 3-in-1 printer with 3D printing, CNC carving and laser engraving capabilities, while the Snapmaker J1S is an Independent Dual Extruder (IDEX) 3D printer focused on high-speed, high-resolution printing. Both printers offer a large build volume, with $300 \times 200 \times 200$ mm for the J1S and $350 \times 400 \times 400$ mm for the Artisan. They also feature a heated bed and an enclosure, which allows for a wide variety of filament types to be used in the project.

3.1.2 Models

The 3D models used for the forearm and hand components were derived from the open-source InMoov project by Gael Langevin. Some models were adopted directly, while others were modified or replaced with community remixes to enhance mechanical stability. For example, a remix that merges the two-piece forearm shell into a single unit was utilized to increase rigidity, made feasible by the large build volume of the Snapmaker J1S printer[11]. Additionally, a custom stand was designed in Autodesk Fusion 360 to secure the arm vertically and house electronic components.

3.1.3 Filament

Based on forum discussions and recommendations from the InMoov website, two filament types were considered: Acrylonitrile Butadiene Styrene (ABS) and Polylactic Acid (PLA). ABS offers good heat resistance and durability and it can be repaired using acetone. However, it emits potentially harmful fumes and requires a high bed temperature to prevent warping.

Ultimately, PLA was chosen as the more suitable alternative. It prints at lower temperatures and does not require a heated chamber, making it a better fit for the Snapmaker printers. PLA was also the primary filament type already in use by the MINKT laboratory staff. For this build, PLA from PolyTerra in charcoal black and cotton white was selected.

3.1.4 Settings

Printer settings were calibrated through iterative test runs. Optimal values for temperature, speed and infill were determined to balance print quality with structural integrity. These settings ensured the precise fabrication of finger joints, tendon channels and mounting points, all of which were essential for proper functionality. Although the print settings provided by the InMoov project were taken into account, the results of initial test prints revealed a lack of detail and stability, making further adjustments necessary.

3.2 Hardware

3.2.1 Arduino Uno Rev3

The Arduino Uno Rev3 serves as the processing unit of the project. It samples the sEMG data, processes the signal and converts it into motor control commands. Paired with a Sensor Shield V5.0, it simplifies the connection of sensors and actuators (Figure 1). The shield exposes all standard Arduino input and output pins, each accompanied by dedicated power and ground lines. Additionally, it includes an auxiliary power connector that allows a separate power supply for digital pins D0 to D13. This is an essential feature for powering high-torque servo motors without overloading the Arduino itself. This removes any need for further development of any hardware circuits to connect and power the servo motors. This would have been needed if other microcontroller systems like the Raspberry Pi or the ESP32 would have been used for the project.

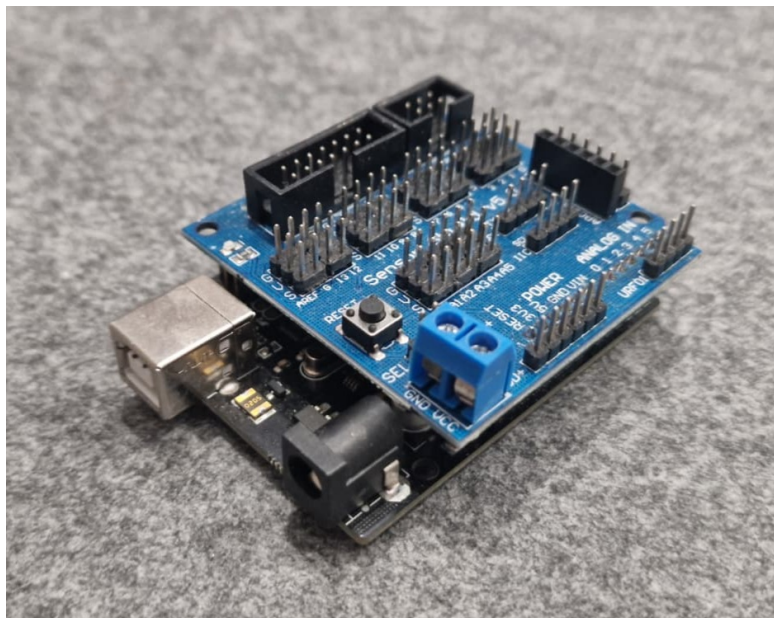


Figure 1: Arduino Uno Rev3 with the Sensor Shield V5.0 attached.

3.2.2 Gravity: Analog EMG Sensor

The Gravity: Analog EMG Sensor [12] captures motor unit action potentials (MUAPs) from the forearm muscles and converts them into an analog signal, which is then sampled by the Arduino. The sensor consists of a dry electrode board and a signal amplification board, as shown in Figure 2a.



(a) Top view of the sensor modules.

(b) Bottom view of the dry electrode board.

Figure 2: Dry electrode board and the signal transmitting board of the Gravity: Analog EMG Sensor.

The dry electrode board (Figure 2b) features three 22 mm $\times$ 35 mm dry electrodes. These are connected to the amplification board via an audio cable and PJ-342 jack. The central electrode acts as the reference, while the two outer electrodes serve as detection electrodes in a bipolar configuration. The electrode board is placed onto the flexor carpi radialis and held in place with an elastic strap. According to the manufacturer, the dry electrodes do not require conductive gel.

The amplification board amplifies input signals within a ± 1.5 mV range by a factor of 1000. It uses a differential input and analog filter circuit. However, the manufacturer does not provide further technical specifications for these two boards. The sensor outputs an analog signal ranging from 0 to 3 V, centered around a 1.5 V reference voltage. The amplification board requires a supply voltage between 3.3 V and 5 V. Power, ground and the analog output are routed to the control unit via a PH2.0-3P connector.

Alternative sensors were not considered viable, as they relied on adhesive electrodes. While such electrodes can provide superior signal quality, their use entails higher operational costs and increased setup complexity.

3.2.3 Micro Digital MG996R Servo Motor

The originally recommended servo motors for the project, the HobbyKing HK15298B, were unavailable from online retailers during the assembly phase. As a result, Micro Digital MG996R servo motors were used as an alternative. The MG996R is a high-torque digital servo with metal gears, capable of delivering a stall torque of 9.4 kgf·cm and an angular velocity of 353 degrees per second at 4.8 V. It supports an approximate rotational range of 120 degrees and operates with a running current between 500 and 900 mA, peaking at up to 2.5 A under stall conditions [13].

3.2.4 Fishing Line

Each fingertip of the bionic arm is actuated using braided fishing line, selected for its durability and non-elastic characteristics as recommended by the InMoov project. Two lines are knotted at the tip of each finger and routed through designated support holes within the fingers, palm and wrist structure. Both lines are anchored to a servo pulley mounted on the motor shaft of the servo motor. Depending on the direction of rotation, the servo motor retracts one of the two lines, thereby inducing either flexion or extension of the finger.

3.2.5 Power Source

To meet the system's power demands, a power supply capable of delivering at least 15 A was required. This covers a worst-case scenario where all five servo motors stall simultaneously, drawing around 12.5 A in total. Initial tests with the Siglent SPD3303C (5 A limit) proved insufficient, while the QJE PS3010 (10 A max) allowed for testing the motors simultaneously. For reliable operation under peak loads, a dedicated 15 A power supply was chosen for the final setup. This unit powers only the servo motors, while the microcontroller and sensors are typically supplied via USB during development. The final stand includes a DC port and an integrated on/off switch.

3.3 Software

3.3.1 Sampling

The analog signal from the sensor unit is digitized using the Arduino's 10-bit analog-to-digital converter (ADC). The EMG sensor library includes example scripts supporting sampling rates of 500 Hz and 1000 Hz, which served as the basis for the project's code [14]. A sampling rate of 1000 Hz was chosen, as it satisfies the Nyquist-Shannon sampling theorem for the expected sEMG frequency range of 0 to 500 Hz.

3.3.2 Filtering

As mentioned in the previous section, initial tests were performed using the example code provided by DFRobot [14]. This code includes a digital filtering stage designed to suppress power line interference and artifacts not addressed by the sensor’s analog circuitry. The filter consists of a cascaded structure: a fourth-order notch filter, a second-order Butterworth high-pass filter and a second-order Butterworth low-pass filter, forming an overall band-pass response. The notch filter can be configured for either 50 Hz or 60 Hz, depending on regional power line frequency, while the combined filters define a passband between 20 Hz and 150 Hz. The high-pass filter mitigates baseline drift and motion artifacts and the low-pass filter reduces high-frequency noise. This configuration will be referred to as the *Original Filter*.

However, test results revealed residual interference near 50 Hz and 150 Hz, as well as insufficient suppression outside the target frequency range for muscle activity. These issues indicated that power line noise and its harmonics remained in the signal, increasing the likelihood of false detections. To address this, two custom digital filters were implemented to evaluate whether signal quality and muscle onset detection could be improved.

The implemented filters are based on the same principle demonstrated in the provided sample code, with filter coefficients computed using the SciPy Python library [15] [16]. Both filters employ a passband ranging from 60 Hz to 135 Hz, a range chosen to suppress powerline interference while preserving the primary frequency components of the sEMG signal. The filters were evaluated according to the methodology outlined in Section 3.4.2 and further assessed through feature extraction analysis. This evaluation aims to determine whether enhanced filtering performance leads to improved detection accuracy.

3.3.3 Feature Extraction

Feature extraction reduces the dimensionality of the processed sEMG signal while preserving essential characteristics required for control decisions. In direct control algorithms, time-domain features are often favored due to their computational efficiency and suitability for real-time implementation.

As mentioned in section 2.6 Cho et al. systematically evaluated various time-domain features for muscle activity onset detection, including Root Mean Square (RMS), Variance (VAR) and Waveform Length (WL). These three features are implemented and tested for this system. For this project, detection accuracy is prioritized over any other metrics.

3.3.4 Control Algorithm

A direct control algorithm operates the bionic hand. Control is binary (open/close) and is triggered by voluntary muscle contraction. While reproducing complex gestures from sEMG is possible [17], [18], it requires more advanced processing and a multi-sensor setup which is outside the scope of this project.

The runtime pipeline follows section 3.3: sampling at 1000 Hz (section 3.3.1), digital band-pass filtering (section 3.3.2) and time-domain feature computation (section 3.3.3). The filter used by the algorithm is determined by the methods explained in section 3.4.2. The optimal feature and threshold were selected through offline evaluation (section 3.4.3) using a labeled dataset (section 3.4.1) to maximize accuracy while minimizing detection delay.

Runtime loop. At each iteration, the Arduino:

1. acquires one sEMG sample,
2. updates the band-pass filter state,
3. computes the selected feature over a sliding window,
4. compares it to the decision threshold,
5. closes the hand if the threshold is exceeded and opens it otherwise.

3.4 Evaluation and Testing

3.4.1 EMG Recording and Labeling Protocol

To perform the evaluations described in the following sections, a data set was created from four adult volunteers (23–30 years, BMI in the normal range). Each recording (20–60 s) contained a randomized mix of short and long contractions and rest periods. This variability was chosen to avoid biasing the detector toward a fixed cadence or duration and to test robustness across different contraction lengths and timings. The signals were transmitted via serial connection to a laptop and recorded with CoolTerm[19]. Recordings were then manually labeled following a fixed protocol and split *by recording* into disjoint training and test sets to prevent temporal leakage. The training set was used to determine thresholds for the feature evaluation, and the held-out test set was used exclusively for assessment. Labeling, evaluation, and plotting were implemented in Python.

3.4.2 Filter Evaluation

To comprehensively evaluate the performance of each filter configuration (original, 4th-order Butterworth, and 6th-order Butterworth), all analyses were conducted using the same raw sEMG signal as input to ensure direct comparability. This approach eliminated variability caused by differences in muscle activity or electrode placement and allowed the results to reflect only the effects of the filtering methods. The evaluation focused on a set of quantitative measures that capture both signal quality and frequency content, providing an objective basis for comparing the filters.

Time-Domain Analysis

A time-domain plot is used to compare the raw and filtered signals. This should give a visual impression of the filter performance. The time-domain analysis focuses on the following aspects:

- Baseline level and drift
- Noise level during rest segments
- Clarity and repeatability of contraction bursts

Frequency-Domain Analysis

Spectral analysis is used to quantify how well each filter suppresses unwanted components while preserving sEMG content.

The analysis uses the Discrete Fourier Transform (DFT):

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1 \quad (6)$$

where $x(n)$ is the discrete-time signal and N is the total number of samples. The DFT was applied to both the raw and filtered signals, enabling a direct comparison of how each filter configuration affects the signal's frequency spectrum.

Welch's method was used to calculate the Power Spectral Density (PSD) of the signals [20]. This reduces variance in the spectral estimate by dividing signals into K overlapping segments of length L , applying a *Hann* window to each segment:

$$w(n) = 0.5 \left(1 - \cos \left(\frac{2\pi n}{L-1} \right) \right), \quad 0 \leq n < L \quad (7)$$

and then computing the windowed periodogram of each segment via FFT. Let $x_k(n)$ denote the k -th segment, F_s the sampling rate, $N_{\text{fft}} \geq L$ the FFT length, and define the window-energy normalization factor as

$$U = \sum_{n=0}^{L-1} w^2(n). \quad (8)$$

The modified periodogram for segment k at frequency bin m ($f_m = \frac{mF_s}{N_{\text{fft}}}$) is

$$P_k(f_m) = \frac{1}{F_s U} \left| \sum_{n=0}^{L-1} x_k(n) w(n) e^{-j2\pi mn/N_{\text{fft}}} \right|^2. \quad (9)$$

To get a smoothed PSD curve the averaged periodogram is calculated as:

$$P_{xx}(f_m) = \frac{1}{K} \sum_{k=1}^K P_k(f_m). \quad (10)$$

To further quantify filter effectiveness, the total power within defined frequency bands was calculated by integrating the PSD:

$$P_{\text{band}} = \int_{f_1}^{f_2} P_{xx}(f) df \quad (11)$$

The total signal power over the full frequency range was computed as:

$$P_{\text{total}} = \int_{f_{\min}}^{f_{\max}} P_{xx}(f) df \quad (12)$$

From these values, the relative contribution of each frequency band was expressed as a percentage:

$$\text{Percentage} = \left(\frac{P_{\text{band}}}{P_{\text{total}}} \right) \times 100\%. \quad (13)$$

Because the transfer function does not capture sensor noise, ADC artifacts or numerical effects on the microcontroller, we are using PSD-based band-power analysis on real sEMG to quantify how well each filter preserves sEMG energy in the intended passband. The same analysis quantifies suppression outside the passband and enables comparisons across filter designs. These metrics provide objective, implementation-level evidence of filter performance.

3.4.3 Feature Evaluation

Given the project’s emphasis on detection accuracy over latency, three features—RMS, VAR and WL were implemented to evaluate their effectiveness in detecting muscle activation from sEMG signals:

- Root Mean Square (RMS): Effective signal value, representing power.

$$RMS = \sqrt{\frac{1}{N} \sum_{n=1}^N x(n)^2} \quad (14)$$

- Variance (VAR): Dispersion of signal values around the mean μ .

$$VAR = \frac{1}{N} \sum_{n=1}^N (x(n) - \mu)^2 \quad (15)$$

- Waveform Length (WL): Cumulative length, capturing signal complexity.

$$WL = \sum_{n=1}^{N-1} |x(n+1) - x(n)| \quad (16)$$

Where $x(n)$ is the n -th sample and N is the window size.

For each feature, optimal thresholds were determined using a labeled training dataset. Signals were first pre-processed with the filter chosen in Section 3.4.2. Features were then computed with a sliding window, with window sizes from 10 ms to 200 ms in 10 ms increments. For each window size we performed a two-dimensional grid search over the decision threshold. Candidate thresholds were taken as the midpoints between sorted feature values from the two classes to cover the range densely and to avoid placing a threshold exactly on noisy samples. For every (feature, window) configuration we evaluated training accuracy and selected the threshold that maximized performance; when classes were imbalanced we used balanced accuracy and broke ties by lower detection delay. Detection delay was defined as the time between the true contraction onset and the first threshold crossing under the chosen windowing scheme. The selected thresholds were aggregated across training recordings using the median and then applied unchanged to a separate labeled test dataset. For each window size and feature a binary prediction was generated by comparing the computed feature value to the selected threshold.

Two primary metrics were calculated:

- **Accuracy:** The proportion of correctly classified samples (true positives and true negatives) out of the total.
- **Detection Delay:** The average latency (in milliseconds) between the actual onset of a muscle contraction (ground truth) and the first correct detection by the algorithm.

Results were visualized using boxplots, line plots across window sizes and heatmaps. These informed about the final choice of feature and threshold for real-time control.

3.4.4 Finger Movement Test

To test the servo motors, a single-finger prototype called the Finger Starter Kit was printed and used. This kit is a simplified version of the InMoov finger, using the same mechanism used in the final hand. The kit includes a holder for the servo motor, a servo pulley and a finger with two tendon channels. This setup was used to test every finger and motor used in the final hand. It was also used to test different variations of servo pulleys published online to improve finger tension. Each servo was tested individually for proper operation and current draw.

Following these initial tests, servo motors were mounted in the forearm and connected to the fingers via the tendon system. Individual fingers were then actuated to verify tension and smooth movement. A durability test involving 100 repeated flexion-extension cycles was performed to assess mechanical reliability, monitor tendon line wear and identify weaknesses in the 3D-printed components.

3.4.5 Full System Test

A test of the entire system with the chosen filter and features was conducted to see the system flaws. This test should not be done within the normal program of the MINKT laboratory but rather in a controlled environment to see the first flaws in the system.

For this test four subjects of the MINKT laboratory staff were chosen. Each subject independently placed the sensor over his/her flexor carpi radialis using the elastic strap. They then performed random-duration muscle contraction and relaxations, simulating unsupervised interaction typical for the MINKT laboratory setting. For this test a threshold and window size were predetermined from the evaluation done previously. This is done to see if those values are also fitting for every person.

4 Results

4.1 3D Printing

The calibration of the 3D printers, finding the right print and slicing settings and the printing of the initial parts took place over a three week period. The parts of the InMoov project printed as expected. Before the parts can be assembled some needed adjustments, like smoothing of the edges and redrilling the holes according to size. For the assembly of the arm the tutorial on the InMoov website was used [21]. The arm and the remix forearm can be seen in Figure 3.

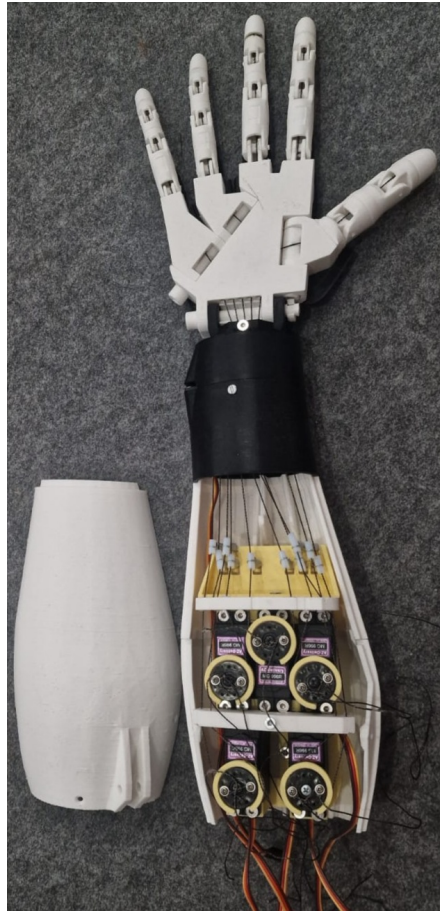


Figure 3: Final version of the bionic arm.

After the finger movement test, wear on the 3D-printed parts where tendon lines pass through was observed. The 3D models were modified to include Teflon tubes at points where the fishing line passes through to increase durability and reduce friction. The Teflon tubes were selected for their low coefficient of friction compared with printed PLA. The holes for the tubes were designed to be press-fit, securing them while still allowing removal (Figure 4).

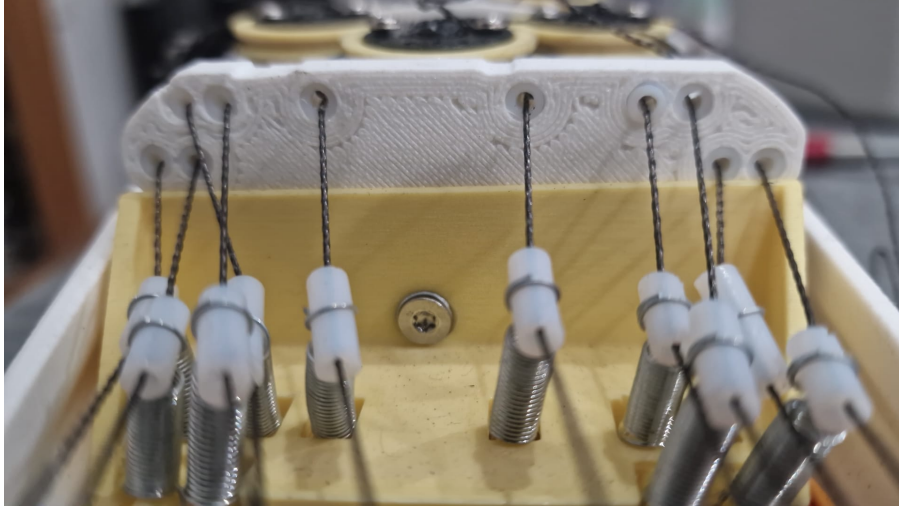


Figure 4: Implementation of the Teflon tubes in the servo bed.

The custom stand was printed in two parts, the base and the top part. The top part integrated the InMoov biceps connector as an attachment point for the arm. The bottom part had to be printed multiple times because the corner of the bottom layer detached. For this an adhesion spray and "Micky-Mouse ears" were used, which are large circles printed around the corners to increase the surface area. To secure the two parts together, M4 heat threads were embedded into the print. The threads are heated using a soldering iron and pressed into the print, which fuses the metal with the PLA. After installing the electronics, some design flaws became apparent. The Arduino holder was placed too close to the on-off switch and several holes were undersized and needed to be cleaned up using sandpaper. Despite these issues, the print was used for the prototype.

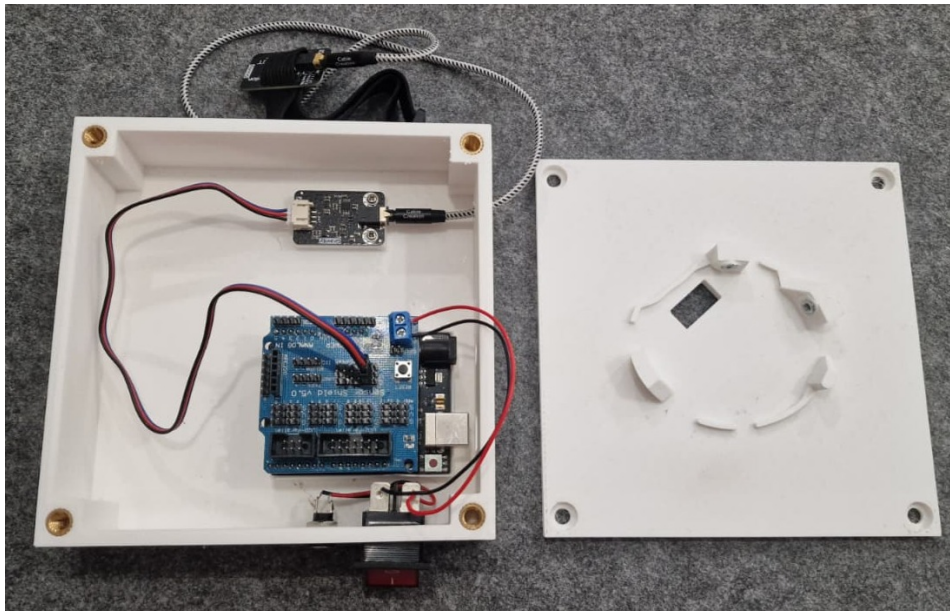


Figure 5: Final version of the custom stand.

4.1.1 Filament and Settings

The final print settings were established using different 3D test prints. The final printing settings and slicing configuration are listed in table 1.

Table 1: 3D printing settings for PolyTerra PLA and slicing configuration.

Print settings			Slicing settings	
Setting	Provided Ranges	Used	Setting	Value
Printing Speed	30–70 mm/s	60 mm/s	Layer Height	0.12 mm
Printing Temperature	190–230 °C	210 °C	Wall Thickness	2 mm
Bed Temperature	25–60 °C	60 °C	Model Structure	Strong
Fan	ON	ON	Support	Tree
			Adhesion	Brim
			Infill	30%
			Infill Pattern	ZigZag

4.2 Hardware

4.2.1 Arduino Uno Rev3

The Arduino Uno Rev3 and Sensor Shield performed reliably throughout testing. No functional issues were encountered during data acquisition, filtering, or motor control tasks. The pin configuration can be seen in Table 2.

Table 2: Arduino Uno Rev3 pin configuration for sEMG sensor and finger servos.

Component	Arduino Pin	Description
sEMG Sensor	A1	Analog input for muscle activity signal
Potentiometer	A2	Analog input for contraction threshold adjustment
LED	D2	Digital output for LED indicator
Pinky Servo	D3	Digital PWM output to pinky finger motor
Ring Servo	D4	Digital PWM output to ring finger motor
Middle Servo	D5	Digital PWM output to middle finger motor
Index Servo	D6	Digital PWM output to index finger motor
Thumb Servo	D7	Digital PWM output to thumb motor

4.2.2 Gravity: Analog EMG Sensor

Surface EMG signals were recorded using the Gravity Analog EMG Sensor positioned over the flexor carpi radialis. The dry electrode setup maintained reliable contact during extended trials. However, when the electrodes were removed and reapplied signal instability occurred, often in the form of high-amplitude transient noise. A stable recording after reapplication could usually be restored by gently pressing the electrode board onto the skin for a few seconds. During these tests the use of conductivity gel was also evaluated to determine whether it could improve signal quality. The results showed no significant improvement making its use unnecessary for the standard setup. Moreover, applying conductivity gel would negate the primary advantages of dry electrodes such as simplified setup and reduced waste compared to adhesive electrodes. Raw signal data displayed a clear DC offset (~ 1.5 V), moderate baseline noise and distinguishable bursts corresponding to voluntary muscle contractions (Figure 6).

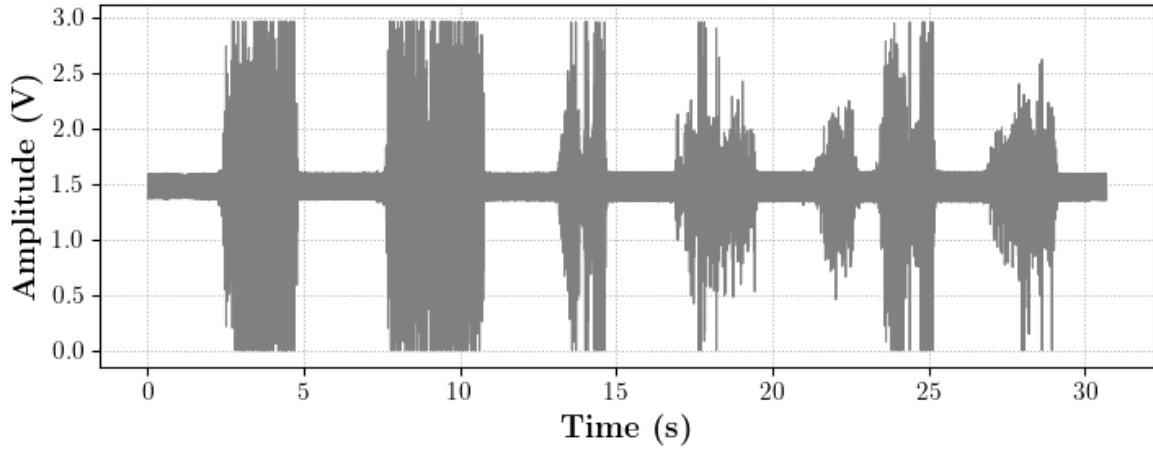


Figure 6: Example segment of a raw sEMG signal acquired from the flexor carpi radialis, showing baseline noise, DC offset, power line interference and muscle contraction periods.

4.2.3 Micro Digital MG996R Servo Motor

The MG996R servo motors were tested individually before mounting. A custom Arduino script verified their range, current draw and functionality. All tested motors passed inspection. They were then mounted onto the servo bed within the forearm structure. Connector cables were routed carefully through the bottom of the bed. This avoided interference with motor movement and tendon lines. Before installing pulleys, motors were set to a 90° angle. Pulleys were then mounted and tension lines connected and tightened. To improve this mounting process different servo pulleys were tested. In the end the original pulley was used because the remixes did not provide any additional benefits.

4.2.4 Fishing Line

Optimizing finger movement required multiple testing rounds. Initial individual finger assessments often revealed inadequate tendon line tension, requiring tightening. The subsequent 100-cycle durability test highlighted performance issues. The middle finger struggled to close fully after repeated cycles. The pinky experienced difficulty extending completely. Incomplete extension was a general issue across all fingers.

5 mm $\times$ 20.5 mm extension springs were incorporated into the servo bed to maintain consistent line tension. Teflon tubes were integrated in the end loops of the springs to minimize friction. The method for securing lines at fingertips was revised. Lines were separated, tied individually and glued to prevent knot shifting. As shown in Table 3 the initial servo angles were calibrated individually to achieve full extension. Even with those measures the knot of the middle finger opened and slipped through the holes in the finger. To prevent this a M3 nut was placed in the tip of the finger to which the strings were attached. These changes increased the line tension but did not eliminate the necessity for re-tensioning.

Table 3: Servo motor end angles for opening the hand.

Finger	Open Angle (°)	Description
Thumb	93	End angle for fully open thumb position
Index	120	End angle for fully open index finger position
Middle	99	End angle for fully open middle finger position
Ring	110	End angle for fully open ring finger position
Pinky	100	End angle for fully open pinky finger position

4.2.5 Power Source

The cumulative current demand of the servos (up to 12.5 A) exceeded the capacity of initial test power supplies (Siglent SPD3303C: 5 A; QJE PS3010: 10 A). Testing confirmed the necessity of a higher capacity supply. The selected 15 A power supply provided stable operation for all motors simultaneously during subsequent tests.

The power source is connected with the Sensor Shield. By removing the power jumper on the Sensor Shield, the servo power rails are isolated from the Arduino’s 5 V rail. This requires a separate supply for the Arduino and ensuring a common ground between supplies

4.3 Evaluation and Testing

4.3.1 Filter Evaluation

The three filters proposed in the method section were implemented and evaluated with the previously described metrics. The filter provided by DFRobot is labeled as *Original Filter* (blue). The other two filters implemented were named *4th Order Butterworth* (orange) and *6th Order Butterworth* (green).

Time-Domain Analysis

Figure 7 shows the time-domain plots of the raw and filtered signals. The raw signal exhibits a strong DC offset, powerline interference and baseline noise. The original filter attenuates these components but retains some low-frequency drift. Both the 4th-order and 6th-order Butterworth filters eliminate the offset and significantly remove noise outside the passband. The resulting signals are visually very similar, though the 6th-order filter provides marginally sharper transitions and slightly less baseline variation.

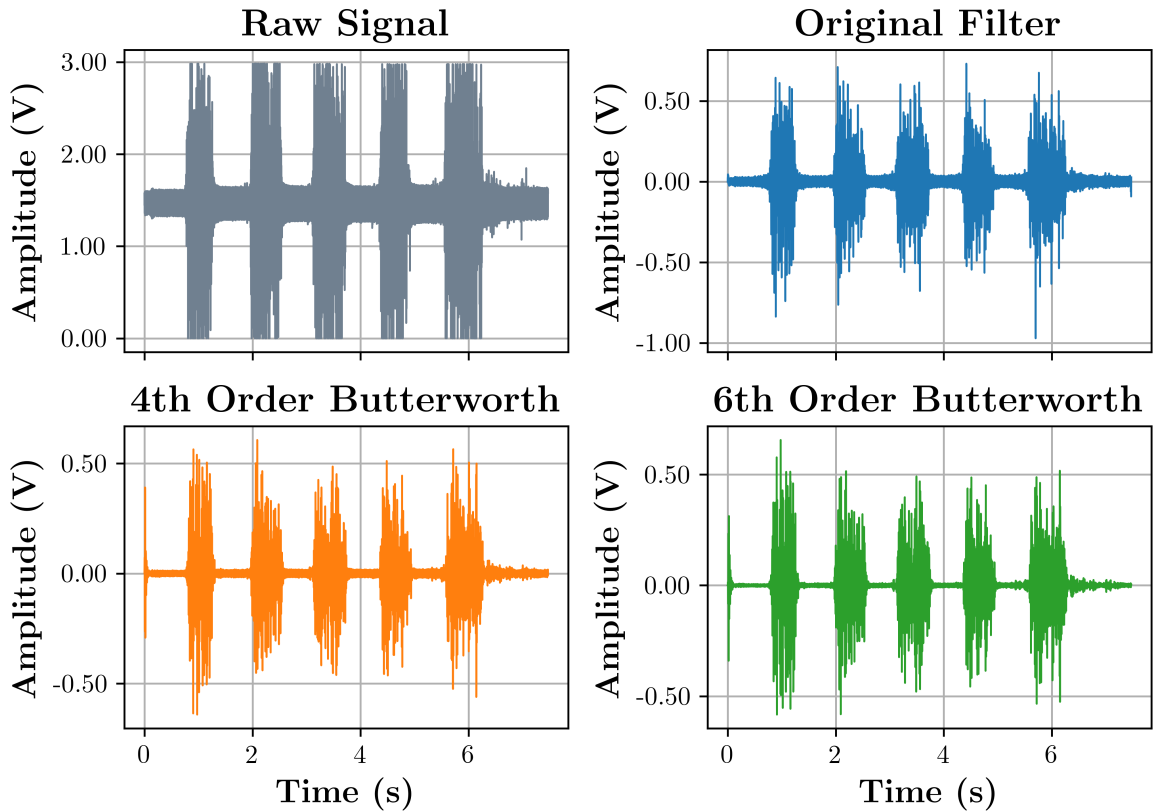


Figure 7: Time-domain comparison of raw and filtered EMG signals.

Frequency-Domain Power Analysis

To evaluate the frequency characteristics of the raw and filtered EMG signals, the power spectral density (PSD) was estimated using Welch’s method. Figure 8 shows the PSDs. The raw signal exhibits broad spectral energy from roughly 50 Hz to 450 Hz with a prominent line near 50 Hz that indicates power-line interference. The Original Filter does not suppress this line effectively. Both Butterworth filters reduce the 50 Hz component and the overall out-of-band power, although a small residual remains.

Within the analysis band used by the controller (65–135 Hz) all filters show similar levels, as expected for unit passband gain. Differences appear in the transition toward the high-frequency stopband: the 4th-order Butterworth begins attenuating earlier but with a shallower slope, whereas beyond about 250 Hz the Original Filter exhibits stronger attenuation. The 6th-order Butterworth shows the steepest high-frequency roll-off and therefore the lowest out-of-band power overall. Hence its advantage is stronger suppression outside the analysis band rather than a change in passband energy.

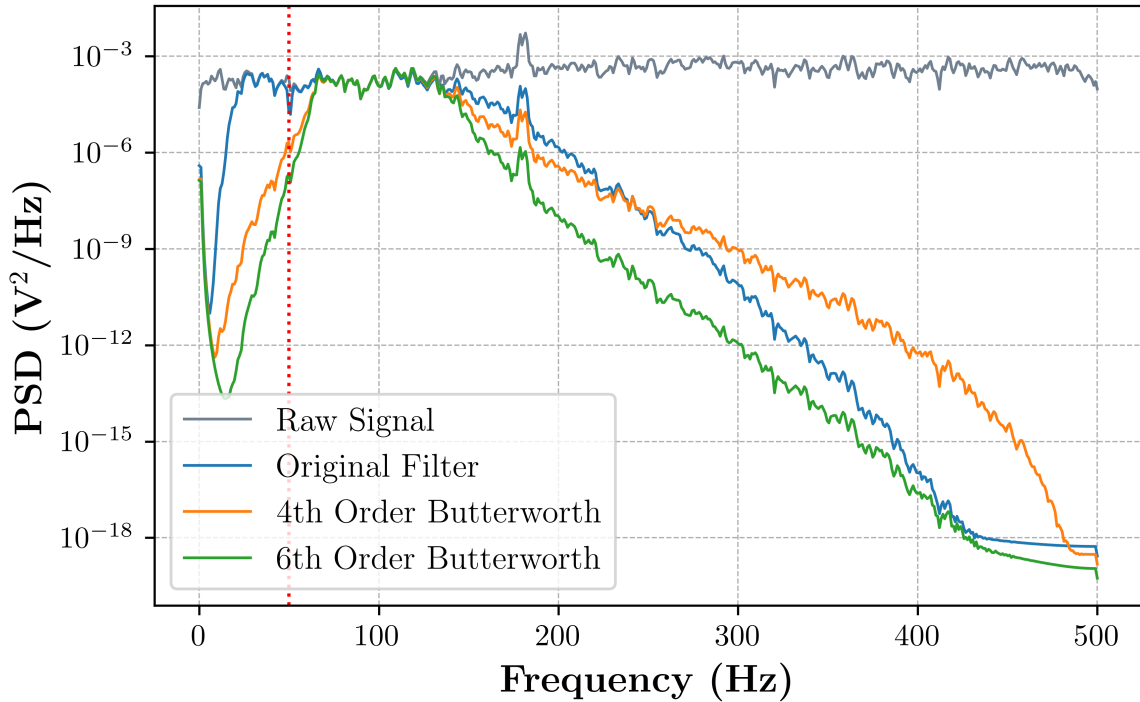


Figure 8: Power spectral density (PSD) of raw and filtered EMG signals.

When total power is split into frequency bands (Figure 9), the original filter increases the share in 65–135 Hz. It still leaves sizable energy below 45 Hz and in 135–200 Hz. Both Butterworth designs shift additional power into 65–135 Hz and suppress out-of-band energy. The 6th-order variant yields the highest concentration in 65–135 Hz with minimal power in adjacent bands and strong attenuation around 50 Hz. The 4th-order

shows the same pattern with slightly weaker suppression. Because EMG can contain physiological content above 200 Hz, we do not label that region as noise. In this thesis we evaluate filters by how well they suppress energy outside the desired passband of 65–135 Hz. On these spectral criteria the 6th-order Butterworth is preferred; in real-time use the 4th-order may be chosen if its lower group delay reduces detection delay.

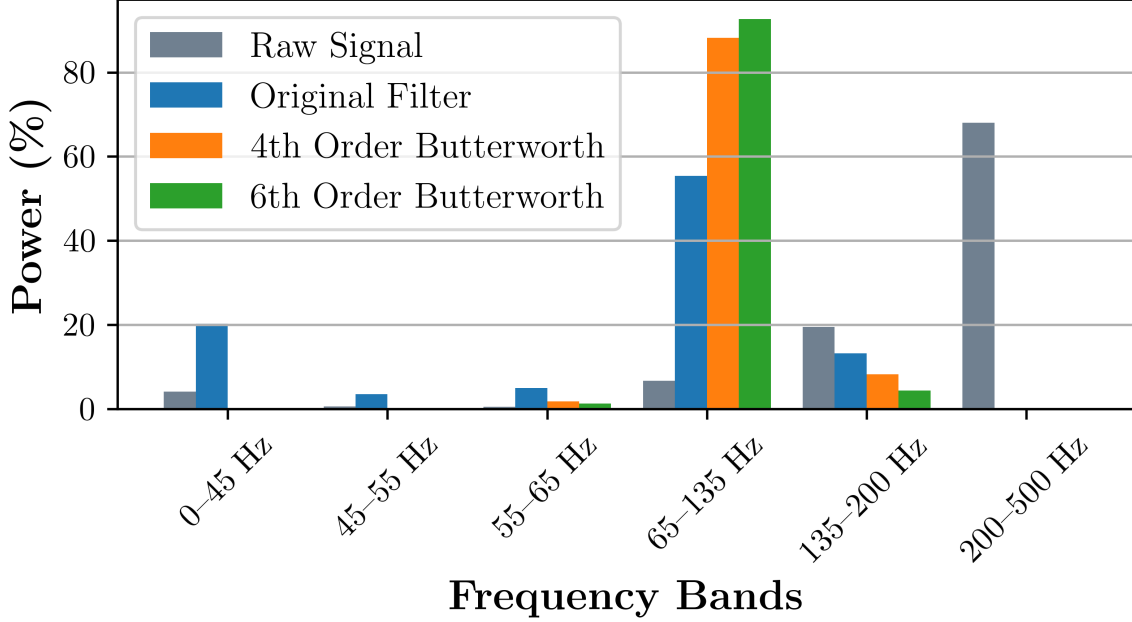


Figure 9: Power distribution across six frequency bands for the raw and filtered EMG signals.

4.3.2 Feature Evaluation

The following figures show the evaluation results obtained using feature-specific thresholds derived from the labeled training dataset (Section 3.4.3). For each feature, the corresponding threshold was used to compute accuracy and detection delay across different window sizes using the test dataset.

In Figure 10 (top), the average classification accuracy for each feature is shown as a function of window size. Accuracy improved for all three features with increasing window size. RMS and WL consistently outperformed VAR across the entire range, with the gap being most visible at smaller windows. For window sizes above 120 ms, RMS and WL reached similar accuracies above 0.955, whereas VAR remained slightly lower but continued to improve gradually with larger windows.

In Figure 10 (bottom), the average detection delay for each feature is plotted against the window size. Detection delay increased for all three features as the window size grew. RMS consistently produced the lowest delays, followed closely by WL, while VAR showed the highest delays throughout. Above 50 ms, RMS and WL differed only marginally in delay, whereas VAR maintained a consistently higher value across the full range of window sizes.

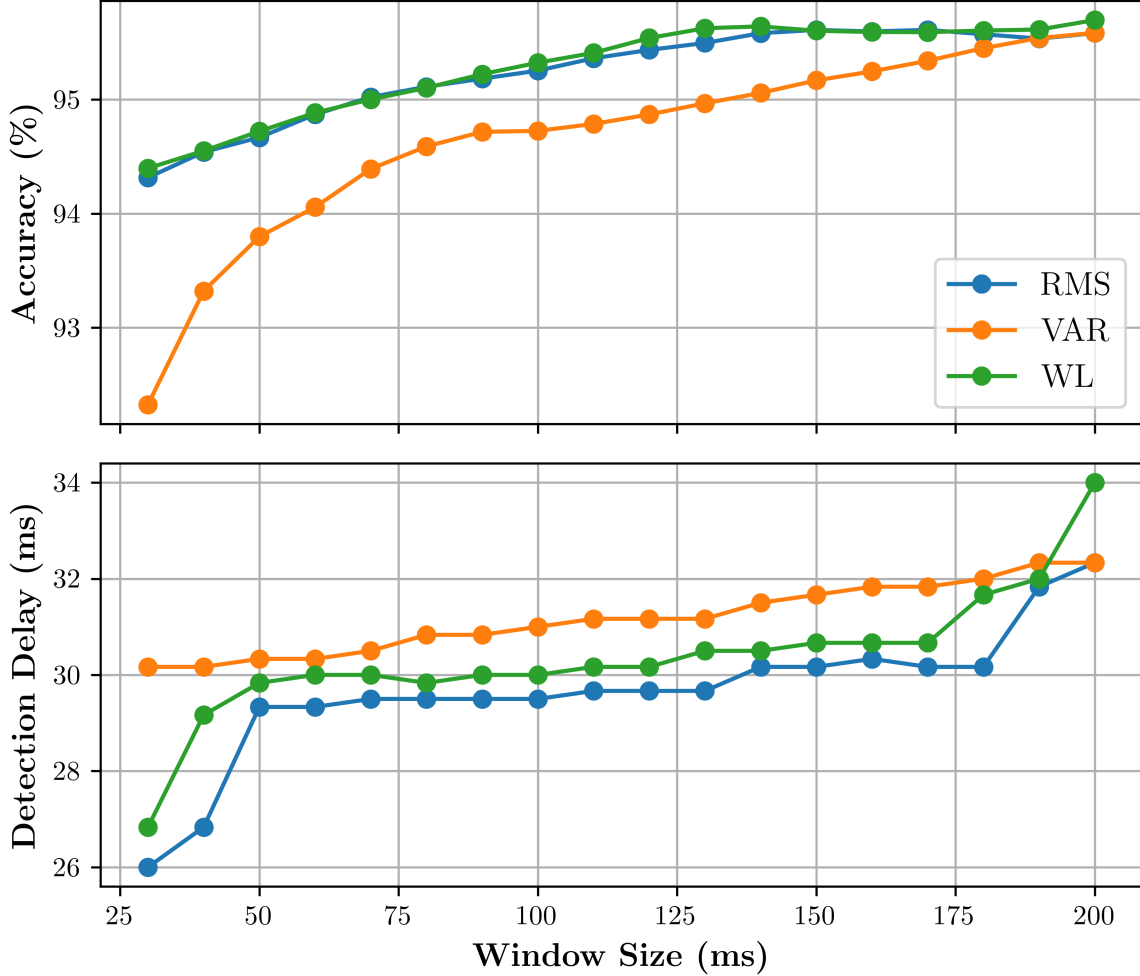


Figure 10: Accuracy and detection delay across window sizes.

Based on the evaluation, RMS and WL demonstrated the highest classification accuracies across all tested window sizes, consistently outperforming VAR. While RMS generally achieved the lowest detection delays, the difference to WL was small, particularly at mid-range window sizes. At a window size of 90 ms, both RMS and WL reached accuracies above 0.951 with delays around 30 ms, representing a favorable trade-off between speed and accuracy.

4.4 Control Algorithm Test

For the control algorithm test, all three features were implemented in an Arduino script using the 6th-order Butterworth filter. The thresholds were set to the values obtained from the training dataset (Section 3.4.3). The algorithm was evaluated with a window size of 90 ms. Figure 11 presents the real-time feature extraction results for RMS, VAR and WL during alternating contraction and relaxation phases, with the vertical red dashed lines marking manually annotated transition points. For all three features, the values consistently crossed their respective thresholds (gray dashed lines) during muscle activation. RMS and WL exhibited clear separation from the threshold in both phases, suggesting that the threshold could be set higher to reduce false positives. In contrast, VAR values frequently fluctuated near the threshold and occasionally crossed it during contraction, making detection less consistent. In this test, RMS and WL both provided stable and reliable identification of the contraction phase.

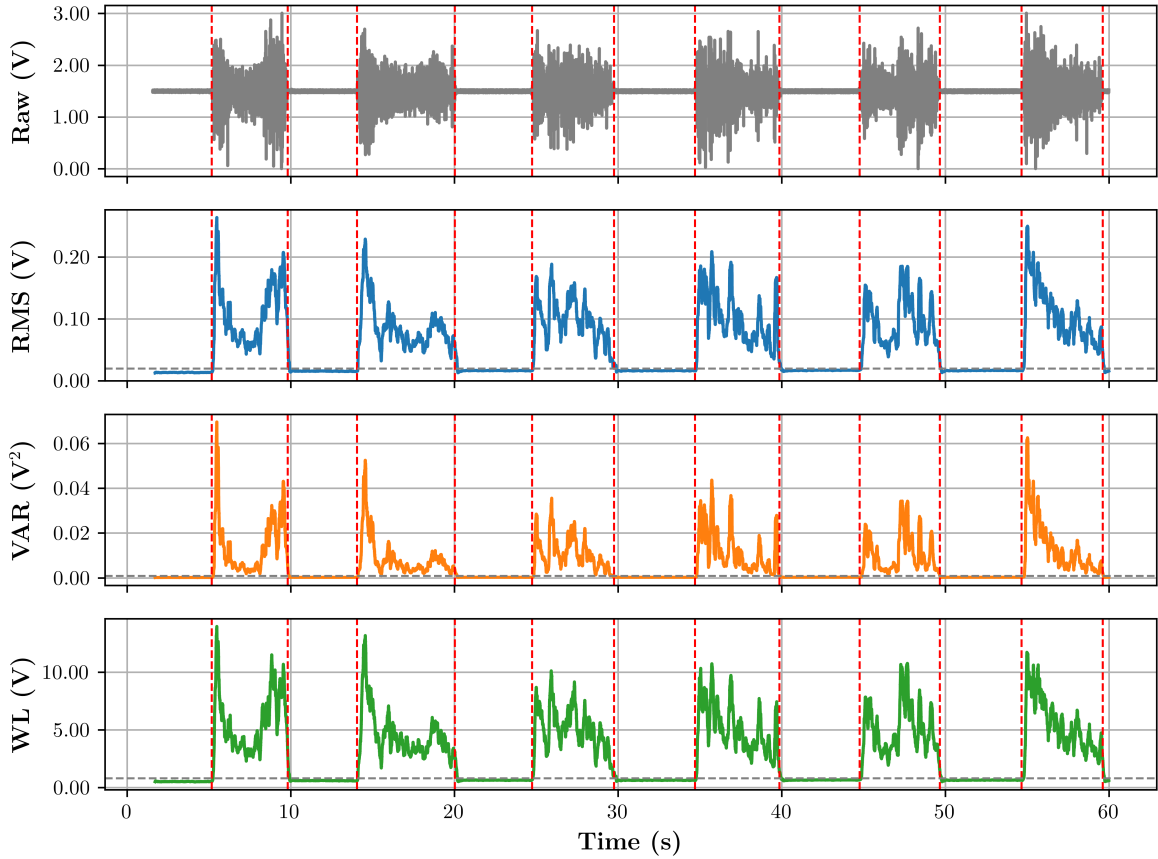


Figure 11: Real-time test of the features with the 6th-order Butterworth filter.

The real-time evaluation confirmed that both RMS and WL provided stable and reliable detection of the contraction phase, with clear separation from their respective thresholds. Although RMS exhibited slightly lower detection delay in the offline analy-

sis (Section 3.4.3), this advantage was not visibly pronounced during the real-time test. Similarly, WL showed comparable detection performance and threshold stability. As no decisive performance gap was observed between the two features in this setting, both remain suitable candidates for real-time control and the final choice may depend on application-specific factors or further long-term testing under varied conditions.

4.4.1 Full System Test

The complete system was evaluated with four subjects to assess both the control algorithm and the mechanical actuation. In addition, a separate robustness test was conducted to emulate public-lab use, where frequent donning and doffing of the sensor is expected. During the handling test the dry-electrode board was intentionally detached and reattached by four volunteers to evaluate system behavior. Between participants the electrodes and contact surfaces were disinfected and allowed to dry before reuse. The same hygiene procedure will be applied during deployment in the MINKT laboratory.

A recurring issue was observed during these handling trials. Removing the electrode board (or partially reseating the audio jack between the electrode and transmitter boards) intermittently drove the front-end into a high-amplitude state. Electrically, an open or intermittently open input leaves the instrumentation amplifier at very high impedance, making it prone to bias drift, mains pickup and saturation, which appears as large spikes and a sustained elevation of time-domain features. In these episodes the classifier interpreted the signal as continuous contraction and the hand closed unintentionally, sometimes resulting in rapid open/close jitter. Peak current draw up to 9 A was previously observed in such states during early integrated tests.

To mitigate this failure mode we introduced a secondary “disconnection threshold” (T_{disc}) in addition to the normal contraction threshold. T_{disc} was set empirically by recording feature values during intentional disconnection; it was chosen sufficiently above the maximum feature values observed during normal contractions so as not to affect routine control. When the feature exceeds T_{disc} , the Arduino disables servo commands, drives the hand to the open position and illuminates an LED on the stand to indicate a handling fault.

Real-time testing also showed that the primary decision threshold derived offline sometimes required upward adjustment between users and sessions. To allow fast calibration, a rotary potentiometer on the stand provides small online adjustments of the primary threshold, with a software-enforced minimum to prevent inadvertent deactivation.

With the dual-threshold safeguard and user-adjustable primary threshold, the system avoided unintended motion during deliberate disconnection events and recovered normally after reattachment. These observations support the use of a high-level disconnection guard as an implementation-level safety feature for public-lab operation.

5 Discussion

The development and evaluation of the bionic arm system demonstrated the importance of integrating mechanical, electrical and signal processing optimizations to achieve reliable performance.

From a mechanical perspective, the 3D printing process produced structurally sound components, though several refinements were required before assembly. Edge smoothing, re-drilling and the addition of Teflon tubing at tendon contact points improved both fit and durability. The Teflon modification, with its low friction coefficient, is expected to extend tendon lifespan and reduce actuation resistance. The custom stand design met functional requirements but revealed dimensional limitations, such as the Arduino holder’s proximity to the power switch and undersized mounting holes.

Hardware testing confirmed the robustness of the Arduino Uno Rev3 and sensor shield under continuous operation. Servo motor calibration ensured full finger extension, though tendon tensioning remained a recurring challenge. The addition of extension springs and Teflon-lined loops reduced friction, but some re-tensioning was still necessary after prolonged use. Knot security was improved by attaching the tendon lines to an embedded M3 nut at the fingertip, preventing slippage. Power supply limitations emerged during early testing, as simultaneous servo actuation exceeded the capacity of initial units. Upgrading to a 15 A supply and modifying the servo shield to separate Arduino power ensured stable operation, while introducing the need for two power supplies.

Signal acquisition was generally stable with the Gravity Analog EMG Sensor, but electrode removal and reapplication frequently introduced transient high-amplitude noise. This issue persisted into the full system tests, where it was traced to contact inconsistencies and mechanical connection problems within the OYMotion sensor unit, specifically incomplete connection of the audio jack between electrode and transmitter boards. These artifacts caused the control algorithm to misinterpret disconnection events as sustained contractions, leading to unintended hand closures, rapid open/close jittering and current spikes up to 9 A. Addressing this required a two-stage control threshold: a high “disconnection threshold” to detect and suppress false activations and a user-adjustable contraction threshold to adapt sensitivity in real time. This dual-threshold system improved safety, reduced mechanical stress and enhanced operational stability across different users and sessions.

In the filtering evaluation, the 6th-order Butterworth filter provided the most effective suppression of out-of-band noise while maintaining the majority of signal energy in the target 65-135 Hz band. RMS and WL consistently outperformed VAR in accuracy, particularly at smaller window sizes, with RMS generally achieving the lowest detection delays. However, real-time testing revealed that RMS’s delay advantage was negligible in practice and both features maintained clear threshold separation during contractions.

After seeing the results, it was decided to look further into the correlation between the WL and RMS feature and the underperformance of the VAR feature. For our calculation we are using a window of W samples $\{x[n]\}_{n=1}^W$.

The RMS is defined as:

$$\text{RMS} = \sqrt{\frac{1}{W} \sum_{n=1}^W x(n)^2} \quad \Rightarrow \quad \text{RMS}^2 = \frac{1}{W} \sum_{n=1}^W x(n)^2 \quad (17)$$

and VAR where μ denotes the *window mean* $\mu = \frac{1}{W} \sum_{n=1}^W x(n)$:

$$\text{VAR} = \frac{1}{W} \sum_{n=1}^W (x(n) - \mu)^2 \quad (18)$$

$$= \frac{1}{W} \sum_{n=1}^W (x(n)^2 - 2 \cdot x(n) \cdot \mu + \mu^2) \quad (19)$$

$$= \underbrace{\frac{1}{W} \sum_{n=1}^W x(n)^2}_{\text{RMS}^2} - \frac{2\mu}{W} \sum_{n=1}^W x(n) + \frac{1}{W} \sum_{n=1}^W \mu^2 \quad (20)$$

$$= \text{RMS}^2 - 2\mu \cdot \mu + \mu^2 = \boxed{\text{RMS}^2 - \mu^2}. \quad (21)$$

After band-pass filtering, the *true* mean of sEMG within a short window is close to zero, so subtracting μ offers little benefit for amplitude detection. However, μ is a *noisy estimate* computed from only W samples, and its fluctuations enter VAR through the $-\mu^2$ term. This adds window-to-window variability to VAR (especially for small W), which destabilizes threshold crossings and degrades accuracy and detection delay relative to RMS, which does not subtract μ .

We also consider WL:

$$\text{WL} = \sum_{n=2}^W |x(n) - x(n-1)|. \quad (22)$$

WL grows with both signal amplitude and effective bandwidth. With the bandwidth tightly limited by filtering (e.g., 65–135 Hz), the bandwidth factor is approximately constant, so WL scales mainly with the amplitude envelope that also drives RMS. Consequently, WL and RMS become strongly correlated on band-limited EMG and tend to yield similar accuracy and delay, as observed.

The algebra above shows that $\text{VAR} = \text{RMS}^2 - \mu^2$, so any noise in the window mean μ directly increases VAR’s variability. RMS avoids this and therefore provided more stable decisions. WL tracks amplitude under fixed bandwidth, which explains its high correlation with RMS and their similar performance.

Given that the EMG was tightly band-limited to 65–135 Hz, RMS and WL both quantified the amplitude envelope and yielded similar accuracy and delay. To maximize stability against residual spikes while keeping computation simple on the microcontroller, we selected RMS (implemented as mean square to avoid the square root) as the primary feature and retained WL as a validated alternative. The optimal window size of 90 ms offered a strong balance between accuracy (>0.951) and low detection delay (30 ms), aligning with trade-off considerations reported in prior EMG control literature.

Despite successful integration of these mechanical and signal processing improvements, the system has limitations. Testing was limited to four participants in controlled conditions and only one type of dry electrode sensor was evaluated. Tendon re-tensioning remains necessary over extended operation and long-term durability under continuous daily use was not assessed.

Future work should focus on refining mechanical reliability, particularly in tendon tension retention and developing automated recalibration protocols to compensate for electrode reconnection events. Broader testing with varied sensors, larger subject pools and real-world use cases will be critical for validating system robustness.

6 Conclusion

This thesis presented the design, implementation and evaluation of a 3D-printed, EMG-controlled bionic arm. The project integrated mechanical fabrication, sensor signal processing and real-time control algorithms into a functional prototype. Through iterative hardware and software development, key challenges in mechanical durability, signal reliability and control stability were addressed.

Mechanical optimizations, including Teflon tubing at tendon contact points and improved tendon anchoring, reduced friction and enhanced durability, while a custom stand provided stable system integration. Signal processing evaluation identified the 6th-order Butterworth filter as the most effective for isolating EMG activity in the target frequency band. Feature analysis showed that both RMS and WL offered high accuracy and low detection delay, with no decisive performance gap, justifying their inclusion in real-time testing.

Control robustness was improved through a dual-threshold strategy, effectively mitigating false activations caused by electrode disconnection and allowing dynamic user adjustment of contraction sensitivity. These measures resulted in more stable operation across users and sessions.

The prototype achieved reliable real-time hand control in a controlled test environment with four participants. Nevertheless, performance in long-term or unconstrained settings, mechanical tendon stability over extended use and compatibility with alternative EMG sensors remain open areas for investigation.

Future work should focus on extended real-world testing, exploring hybrid feature sets combining RMS and WL and developing automated recalibration methods for electrode reconnection events. Such advancements would further enhance the reliability, adaptability and usability of EMG-controlled prosthetic systems.

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